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Original Contribution

The effects of sodium-glucose transporter 2 inhibition on cardiac surgery-associated acute kidney injury: An open-label randomized pilot study

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HIGHLIGHTS

- This pilot trial showed that empagliflozin reduced the AKI incidence.
- Perioperative empagliflozin did not result in ketoacidosis.
- Findings warrant validation in large, double-blind, placebo-controlled trial.

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ABSTRACT

Background: Sodium-glucose transporter-2 (SGLT2) inhibitors reduced the incidence of acute kidney injury in large cardiovascular outcome trials in patients with chronic heart and kidney failure. Acute kidney injury is a common complication following cardiac surgery. We hypothesized that perioperative SGLT2 inhibition could reduce kidney injury after cardiac surgery, measured with the biomarker neutrophil gelatinase-associated (NGAL).

Methods: In this open-label phase IV, randomized, parallel-group, pilot study, adult patients undergoing elective cardiac surgery with cardiopulmonary bypass were randomized to receive either an SGLT2 inhibitor, empagliflozin (10 mg; oral) once daily, from three days before surgery until postoperative day two, or standard-of-care. The primary outcome was the between-group difference of serum NGAL on the second postoperative day. Moreover, other biomarkers for acute kidney injury were measured, including serum kidney injury molecule-1 (KIM-1), hypoxia-inducible factor-1 alpha (HIF-1α), and urine NGAL/Creatinine and KIM-1/Creatinine ratios. Additional outcomes included acute kidney injury incidence within the first seven days following cardiac surgery according to Kidney Disease: Improving Global Outcomes criteria and metabolic parameters, including ketone body concentrations and glycemic control.

Abbreviations: AKI, Acute kidney injury; CI, Confidence interval; CKD, Chronic kidney disease; CPB, Cardiopulmonary bypass; CSA-AKI, Cardiac surgery-associated acute kidney injury; eGFR, estimated glomerular filtration rate; HIF1-α, Hypoxia-inducible factor-1 alpha; IU, International units; KIM-1, Kidney injury molecule 1; NGAL, Neutrophil gelatinase-associated lipocalin; KDIGO, Kidney Disease: Improving Global Outcomes; SGLT2, Sodium-glucose transporter-2; T1D, Type 1 diabetes; T2D, Type 2 diabetes.

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Results: Between March 2022 and April 2023, 55 patients were included (sex: 73 % male, age: 66 ± 10 years, BMI: 28 ± 4 kg/m^2, empagliflozin n = 25, control n = 30) in the intention-to-treat analysis. There were no significant between-group differences in serum and urine NGAL or KIM-1. However, empagliflozin significantly reduced the incidence of acute kidney injury (20 % vs 66.7 %; absolute difference 46.7 %, 95 % CI, −69.7 – -23.6; $P < .001$). A significant increase in serum HIF-1α after surgery was solely observed in the control group. We observed no between-group differences in the incidence of (euglycemic) ketoacidosis or hypoglycemic events.

Conclusions: In this pilot study, perioperative SGLT2 inhibition was not associated with lower NGAL levels. We observed that SGLT2 inhibition reduced the incidence of acute kidney injury in this small study population. As the results of this pilot study are hypotheses-generating, further validation is needed in a large-scale, double-blind, placebo-controlled, randomized trial, which is currently ongoing.

1. Introduction

Acute kidney injury (AKI) is a common complication following cardiac surgery, with a reported incidence between 20 and 70 %. [1] Cardiac surgery-associated (CSA)-AKI is associated with postoperative morbidity and mortality, prolonged hospital stays, and increased healthcare costs. [2,3] In addition, CSA-AKI can have long-term consequences, including the increased risk of developing chronic kidney disease (CKD) and reduced life expectancy compared to patients without CSA-AKI. [4–7]

The incidence of AKI was consistently lower during follow-up in patients receiving SGLT2 inhibitors for chronic heart and kidney failure, with and without type 2 diabetes. [8–11] In humans, perioperative SGLT2 inhibition around cardiac surgery for AKI prevention has not been investigated. Still, preclinical research in rabbits with AKI after cardiopulmonary bypass (CPB) showed that dapagliflozin reduced serum creatinine and improved the tubular injury score. [12]

SGLT2 inhibitors reduce proximal sodium and glucose re-uptake, leading to an acute reduction in estimated glomerular filtration rate (eGFR) due to activation of tubuloglomerular feedback. This reduction in eGFR leads to a decreased kidney workload. Lower filtered sodium load can reduce kidney oxygen consumption, while SGLT2-induced insulin sensitivity and substrate metabolism improvements may further reduce kidney oxidative requirements. [13–15] SGLT2 inhibitors may be protective in settings of kidney hypoxia during cardiac surgery, which has been directly reported in clinically relevant large animal models and indirectly by measurement of bladder urinary oxygenation in humans. [16,17] However, SGLT2 inhibitors can induce ketogenesis and are associated with perioperative ketoacidosis caused by the fasted state and surgical stress. [18–20] This serious side effect has prompted the Food and Drug Administration (FDA) to recommend interrupting SGLT2 inhibitors at least three days before major surgery. [18–21] Therefore, we performed an open-label pilot study to evaluate the effects of SGLT2 inhibition on AKI biomarkers, AKI incidence, and the incidence of ketoacidosis and hypoglycemia in individuals undergoing cardiac surgery. We hypothesized that perioperative SGLT2 inhibition with empagliflozin would reduce neutrophil gelatinase-associated lipocalin (NGAL), an AKI biomarker.

2. Methods

2.1. Trial design

We performed a prospective, single-center, open-label, phase IV, balanced (1:1), randomized controlled (standard care) pilot study. The study protocol was approved by the medical ethics committee of the Amsterdam UMC and by the Dutch competent authority before the initiation of the trial. The trial protocol was registered at the Dutch Trial Register (https://www.onderzoekmetmensen.nl/en/trial/52118). An independent study monitor validated adherence to the Declaration of Helsinki, good clinical practice guidelines, and quality of data collection. A data safety management board was not involved in the decision-making of this study. All participants signed informed consent before any trial-related activity.

2.2. Study participants

Patients eligible for inclusion were aged between 18 and 90 years and scheduled to undergo elective cardiopulmonary bypass (CPB) assisted cardiac surgery at the Amsterdam University Medical Centre, a tertiary academic hospital in Amsterdam, the Netherlands. Exclusion criteria included prior treatment with SGLT2 inhibitors, type 1 diabetes (T1D) (because SGLT2 inhibitors are not prescribed to patients with T1D due to the increased risk of ketoacidosis), T2D with a body mass index (BMI) below 25 (possibly phenotypically resembling T1D), estimated glomerular filtration rate (eGFR) <30 ml/min, systolic blood pressure below 100 mmHg, a history of ketoacidosis, a known or suspected allergy to the trial product, or individuals who were breastfeeding, intending to become pregnant or not using adequate contraceptive methods.

2.3. Intervention

Participants were randomly assigned to the empagliflozin or control group. Patients in the empagliflozin group received a single dose of empagliflozin 10 mg orally (Boehringer Ingelheim International GmbH via Alliance Healthcare Nederland B.V.) every morning, starting from three days before surgery until two days after surgery. The control group received no intervention and received standard perioperative care. We initially intended to start empagliflozin treatment seven days before surgery; however, due to logistical constraints in the planning of surgery, it became evident that this approach was not feasible. Therefore, we adjusted the number of treatments before surgery in the protocol from seven to three days before recruiting the first participant. Patients with T2D had their preoperative glucose management adjusted, if necessary, under the supervision of an endocrinology consultant, according to the treatment algorithm shown in Supplementary Fig. 1.

2.4. Procedures

After informed consent, patients were randomized, and baseline characteristics were collected. The study workflow is summarized in Supplementary Fig. 2. At the time of inclusion and on the morning of the day of surgery (before the first incision) as well as the first and second morning after surgery (D1 and D2), serum and urine samples were collected and stored to assess biomarkers upon trial completion. Furthermore, arterial blood gases and ketone levels were collected at four specific time points on the day of surgery: before the start of surgery (T1), at time of start of cardiopulmonary bypass (T2), at the end of cardiopulmonary bypass (T3) and at time of transport to the intensive care unit (ICU) (T4). Per standard perioperative care, patients with insulin-dependent T2D received a continuous glucose/insulin infusion preoperatively. In case ketoacidosis (pH <7.3, bicarbonate <18 mmol/l, anion gap >10 mmol/l, and the presence of ketone bodies (Beta-

hydroxybutyric acid >2.0 mmol/l) were suspected, patients were to be treated with a mixed glucose/insulin infusion to suppress ketogenesis and subsequently closely monitored and treated according to local practice. Postoperatively, serum creatinine levels and urinary output were measured according to usual care (creatinine was measured daily, and urinary output was measured hourly until the removal of the urinary catheter upon patients' discharge from ICU). Serum creatinine and urinary output were also evaluated separately to determine if each met Kidney Disease Improving Global Outcomes (KDIGO) 2012 criteria for AKI (Supplementary Table 1). We also recorded glucose levels as measured during routine care (four times daily during ICU admission) and insulin administrations when prescribed by the treating physicians. Adverse events were collected by reviewing in-hospital electronic health records.

2.5. ELISA

Serum and urinary concentrations of NGAL and kidney injury molecule-1 (KIM-1) were measured with, respectively, the LEGEND MAX™ Human NGAL (Lipocalin-2) ELISA Kit (Biolegend, cat.N. 443,407) and the human KIM1 (TIM-1) ELISA Kit (ABCAM, cat.N. AB235081). Serum hypoxia-inducible factor-1 alpha (HIF-1α) levels were detected using the HIF1A Human ELISA Kit (Invitrogen, cat.N. EHIF1A). All measurements were performed after the study concluded and according to the manufacturer's instructions. Urinary levels of NGAL and KIM-1 were normalized for concentrations of creatinine, detected by standard tests at the hospital laboratory of clinical chemistry.

2.6. Outcomes

The primary outcome was the between-group difference of mean postoperative serum NGAL concentrations measured on postoperative day two.

In addition, we calculated the differences in the following biomarker concentrations: postoperative urinary NGAL/creatinine ratios, serum KIM-1 concentrations, urinary KIM-1/creatinine ratios, and serum HIF1-α levels between the first and second postoperative days compared to the day of surgery. Secondary outcomes also included between-group differences of AKI incidence rates according to the KDIGO criteria within the first 7 days following surgery, intraoperative ketone levels, the number of ketoacidosis events (defined as a combination of ketonemia >3 mmol/l in combination with a high-anion gap metabolic acidosis), the mean intraoperative glucose level, the mean peak postoperative glucose level measured in the first 48 h postoperatively, the incidence of hypoglycemia (defined as glucose <4.0 mmol/l), the incidence of hyperglycemia (defined as glucose >10.0 mmol/l) and the incidence of insulin administrations. Lastly, we compared the length of stay in the ICU and postoperative in-hospital complications such as mortality, renal replacement therapy, cerebrovascular accident (CVA), cardiac tamponade, gastrointestinal bleeding, anemia requiring transfusion, atrial fibrillation, other cardiac arrhythmia's, pacemaker implantations, infection of the mediastinum, pneumonia, and delirium.

2.7. Sample size

Based on a previous cohort study that found a difference in mean serum NGAL of 120 ng/ml between patients with and without AKI after cardiac surgery, we defined a between-group difference of 120 ng/ml in serum NGAL concentration on postoperative day 2 as a relevant outcome. [22] To detect a difference of 120 ng/ml, with a standard deviation (SD) of 165 ng/ml, power of 90 %, and alpha at 0.05, we would need 40 patients per group.

In this pilot study, adverse events and ketones were continuously assessed. The between-group difference in AKI incidence was evaluated after 25, 50, and 75 % inclusion. In March 2023, upon 75 % of trial completion, the trial leadership team decided unanimously to cease new patient enrolment for this pilot study based on the notable between-group disparity observed in AKI incidence. Concluding that this clinically relevant outcome was sufficient to warrant confirmation in a large, blinded, and placebo-controlled trial, the medical ethics committee was notified of the trial termination.

2.8. Randomization and blinding

Randomization was performed electronically through the data management system Castor EDC (Ciwit BV, Amsterdam, the Netherlands) using balanced-block randomization, with variable-size random computer-generated blocks of two, four, or six and an allocation ratio of 1:1. By design (open-label study) patients, care providers nor study personnel were blinded to treatment allocation.

2.9. Statistical analyses

We performed all analyses on the intention-to-treat population. When biomarkers were below the detection limit, we assigned the zero value. Discrete data are presented as counts with percentages per group. Continuous variables are presented as mean ± SD or median (interquartile range), depending on the distribution of the data. Within-group comparisons were conducted using the paired Student *t*-test. Between-group comparisons were performed using Fisher's exact test for dichotomous outcomes and unpaired Student's t-test or Mann-Whitney *U* tests for continuous variables, depending on the data distribution. Absolute differences between groups are presented with their corresponding 95 % confidence intervals (95 % CI). Repeated measurements were analyzed using repeated measures ANOVA corrected for multiple measurements between groups (Tukeys), and sphericity was not assumed. To compare the treatment groups, we used the interaction term of the randomization group with time. Additionally, we analyzed postoperative kidney biomarkers by calculating the change in concentration compared to the day of surgery. Statistical analyses were performed using SPSS (IBM version 24; IBM Corp., Armonk, New York).

3. Results

Between March 2022 and March 2023, we enrolled 60 patients planned to undergo cardiac surgery, of which 29 patients were included in the empagliflozin group and 31 in the control group (Supplementary Fig. 3). Before the start of the intervention, five patients dropped out (one withdrew consent, two surgeries were canceled for logistic reasons, and in two cases, the type of surgery was changed from on-pump to off-pump surgery, thereby no longer meeting the inclusion criteria). All remaining patients were included in the intention-to-treat analysis (25 in the empagliflozin group, 30 in the control group). All patients in the intervention group started with empagliflozin three days prior to surgery.

Table 1 summarizes the baseline characteristics and surgery details of all included patients. Overall, the presented patient characteristics were well balanced between groups, except for age, with patients in the empagliflozin group being older. Patients were predominantly older males, overweight, and had normal kidney function at baseline. T2D was present in 5 (9 %) patients; none were treated with insulin therapy before hospital admission.

3.1. Kidney biomarkers

Serum NGAL concentrations on the second postoperative day did not differ significantly between groups, empagliflozin: 70.8 ± 6.0 ng/ml vs. control: 68.5 ± 10.0 ng/ml, with a difference of 2.3 ng/ml (95 % CI, −6.9–2.3; $P = .317$). Other biomarkers for kidney injury, including urinary NGAL and KIM-1 in plasma and urine, did not demonstrate significant between-group differences (Fig. 1). In the control group, HIF-

Table 1
Baseline characteristics of the intention-to-treat population.

	All (n = 55)	Empagliflozin (n = 25)	Control (n = 30)
Age, mean ± SD, years	66 ± 10	69 ± 8	63 ± 10
Male sex, n (%)	40 (73)	18 (72)	22 (73)
Ethnic origin, n (%)			
Caucasian	53 (96)	24 (96)	29 (97)
Other	2 (4)	1 (4)	1 (3)
BMI, mean ± SD, kg/m^2	28.0 ± 4.4	27.6 ± 3.4	28.2 ± 5.1
T2D, n (%)	5 (9)	3 (12)	2 (7)
Diabetes medication, n (%)			
Metformin, n (%)	4 (7)	3 (12)	1 (3)
Sulfonylurea derivate, n (%)	1 (2)	0 (0)	1 (3)
Hypertension, n (%)	28 (51)	13 (52)	15 (50)
ACE-inhibitor, n (%)	9 (16)	5 (20)	4 (13)
Angiotensin 2-receptor antagonist, n (%)	15 (27)	7 (28)	8 (27)
NYHA classification, n (%)			
I	18 (33)	10 (40)	8 (27)
II	28 (51)	12 (48)	16 (53)
III	9 (16)	3 (12)	6 (20)
LVEF, n (%)			
Good >50 %	40 (73)	20 (80)	20 (67)
Moderate 31–50 %	15 (27)	5 (20)	10 (33)
ASA physical status classification, n (%)			
I	1 (2)	0 (0)	1 (3)
II	7 (13)	2 (8)	5 (17)
III	45 (82)	22 (88)	23 (77)
IV	2 (4)	1 (4)	1 (3)
Creatinine clearance, mean ± SD, ml/min	70 ± 16	68 ± 16	73 ± 15
EuroSCORE II, mean ± SD, %	3.0 ± 3.2	2.8 ± 3.4	3.1 ± 3.0
Type of surgery, n (%)			
CABG procedure	3 (6)	1(4)	2 (7)
Single non-CABG procedure	28 (51)	12 (48)	16 (53)
Two or more procedures	24 (44)	12 (48)	12 (40)
Duration of surgery, mean ± SD, min	271 ± 96	249 ± 83	291 ± 104
Duration CPB, mean ± SD, min	150 ± 70	138 ± 60	159 ± 77
Duration cross-clamp time, mean ± SD, min	107 ± 50	100 ± 41	112 ± 56
Intraoperative fluid suppletion, mean ± SD, l	3.4 ± 1.3	3.2 ± 1.2	3.5 ± 1.4
Duration intraoperative hypotension[†], mean ± SD, min	22 ± 28	19 ± 21	24 ± 32

Abbreviations: ACE = Angiotensin-converting enzyme; ASA = American Society of Anesthesiologists; BMI = body mass index; CABG = coronary artery bypass graft; CPB = cardiopulmonary bypass; EuroSCORE = European System for Cardiac Operative Risk Evaluation; IQR = interquartile range; LVEF = left ventricular ejection fraction; NYHA = New York Heart Association; SD = standard deviation; T2D = type 2 diabetes.

[†] Defined as mean arterial pressure < 50 mmHg.

1α increased significantly from 110 ± 152 ng/ml before surgery to 171 ± 182 ng/ml on the first day after surgery (P = .04) and to 198 ± 268 ng/ml on the second day (P = .098). For the empagliflozin group, a change from baseline HIF-1α levels in the postoperative period was not observed (Fig. 1E). A complete overview of all biomarker concentrations is available in Supplementary Table 2.

3.2. Kidney injury

The development of AKI characterized according to KDIGO criteria was observed in 5/25 (20 %) patients in the empagliflozin group and 20/30 (66.7 %) patients in the control group, with a difference of 46.7 % between groups (95 % CI, −69.7 – -23.6; P < .001) (Fig. 2). One patient from each group required postoperative continuous veno-venous hemofiltration due to kidney failure. Creatinine and urine output were evaluated separately per KDIGO criteria and summarized in Supplementary Table 3.

3.3. Metabolic effects

Ketone levels increased during surgery (Fig. 3) in both groups; in 41 (76 %) patients, the peak levels of ketones (1.1 ± 0.6 mmol/l) were measured at the end of CPB. The increase in ketone levels did not differ significantly between the groups. In the empagliflozin group, ketone levels increased by 0.9 ± 0.5 mmol/l compared to 0.75 ± 0.6 mmol/l in the control group, a difference of 0.2 mmol/l (95 % CI, −0.45–0.02; P = .22). There was no significant difference in ketone development between the groups (P value of the interaction term time*group = 0.75). Ketoacidosis was not observed in any of the patients.

The peak glucose concentration during the first 48 h postoperatively was significantly lower in the empagliflozin group (Table 2). Furthermore, the incidence of hyperglycemia and insulin administration due to hyperglycemia were significantly lower in the empagliflozin group than in the control group.

3.4. Adverse events

No significant between-group differences were observed in the incidence of in-hospital postoperative complications (Supplementary Table 4). The median length of stay in the ICU was 23 (19–42) hours in the empagliflozin group compared to 32 (21–47) hours in the control group (P = .096).

4. Discussion

In this open-label, randomized, controlled pilot study, we investigated the effect of SGLT2 inhibition with empagliflozin on AKI biomarkers and AKI incidence. We observed no significant between-group differences in NGAL. However, the AKI incidence, according to KDIGO criteria, was 47 % lower in the empagliflozin group compared to the control group. HIF-1α levels significantly increased postoperatively in the control group, whereas no notable change was observed in the empagliflozin group. Ketone bodies increased significantly during surgery in both groups without an additional effect of empagliflozin. There were no incidences of ketoacidosis or hypoglycemia.

This pilot trial concluded earlier due to a clinical difference observed in AKI incidence. After the trial was stopped, we analyzed biomarkers and, unexpectedly, found no significant between-group difference in NGAL. This was unexpected since we selected NGAL as a sensitive biomarker for AKI detection. [23] We consider several explanations for the discrepancy in our findings. The timing of biomarker collection was potentially suboptimal as we collected the first biomarker samples the morning following surgery. However, the optimal time window is reported to be between 6 and 12 h after surgery. [24] In addition, NGAL is possibly a less sensitive biomarker for detecting the kidney protective effects of SGLT2 inhibition, as a previous study also failed to confirm the renoprotective effects of SGLT2 inhibition with NGAL levels. [25]

The finding that the AKI incidence was lower after SGLT2 inhibition aligns with three meta-analyses of large-scale cardiovascular and kidney outcome trials, reporting lower incidences of AKI during outpatient treatment with SGLT2 inhibition. [8–11] Furthermore, preclinical evidence from a rabbit model of cardiopulmonary bypass-associated AKI demonstrated kidney protective effects of SGLT2 inhibition. [12] So far, results from prospective clinical studies of SGLT2 inhibition in the ICU have not confirmed their hypotheses. In the DARE-19 trial, hospitalized COVID-19 patients with cardiometabolic risk factors were treated with SGLT2 inhibitors. However, this did not result in significant between-group differences in the primary endpoint of organ dysfunction or kidney-specific complications. [26] Similarly, in the DEFENDER trial, adding an SLGT2 inhibitor to the treatment after ICU admission did not improve clinical outcomes despite a (non-significantly) lower incidence

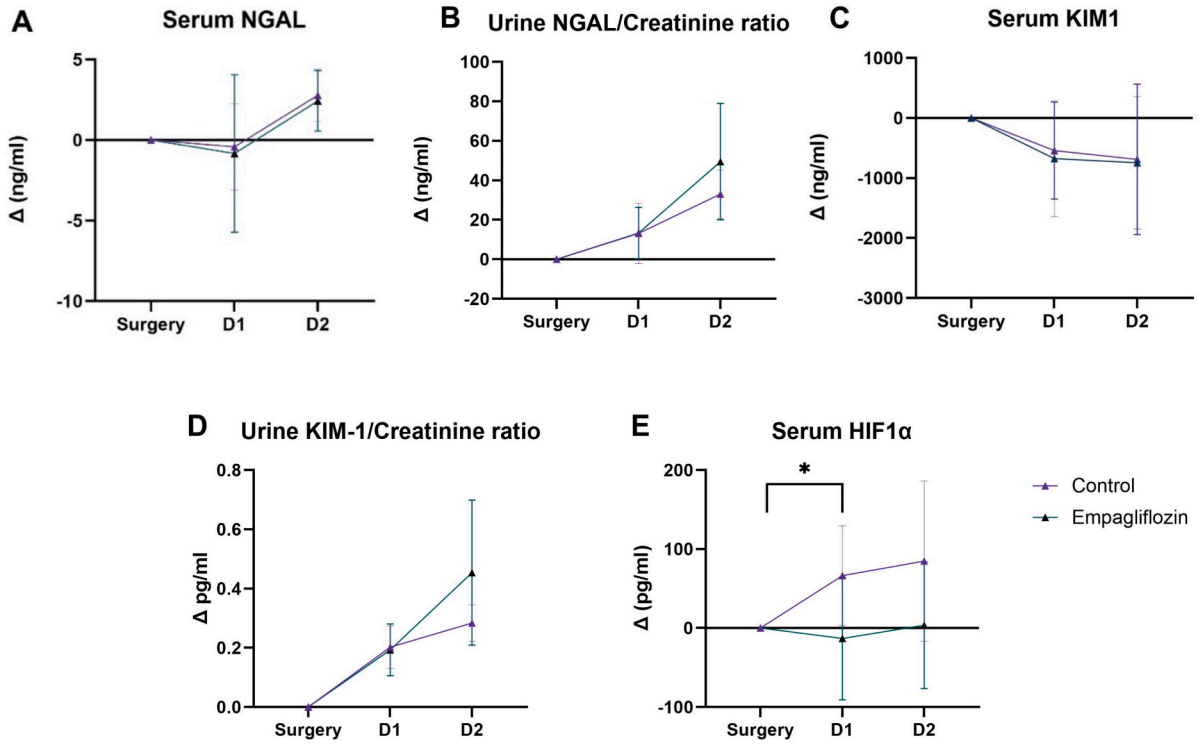


Fig. 1. Differences in postoperative kidney biomarkers concentrations compared to surgery day for the empagliflozin and control group.

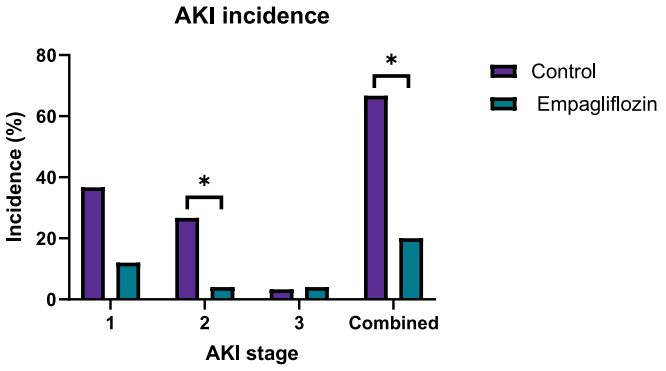


Fig. 2. Incidence rates of Acute Kidney Injury (AKI) across various stages compared between the empagliflozin and control group.

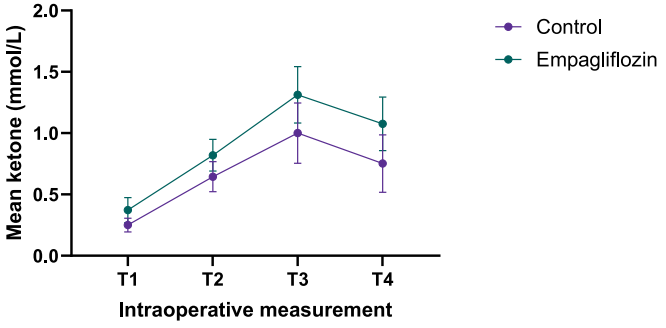


Fig. 3. Intraoperative (mean ± SD) ketone levels.

of kidney replacement therapy. [27] An essential difference between ICU-related AKI and CSA-AKI, such as in our study, is that cardiac surgery allows for the initiation of treatment before the injurious event. [28]

We observed a relatively high AKI incidence of 66.7 % in the control group compared to the incidence rate reported in other studies. However, incidence rates may range from 20 to 70 % depending on the type of surgery, with valve surgery, for instance, being accompanied by a higher risk compared to CABG surgery, definition of AKI used, parameters for AKI used for monitoring, and the study population, with the highest incidence rates reported in Asia. [1,29,30] This study included mainly single-non-CABG procedures (valvular and aortic surgery) and combined procedures associated with a higher AKI risk than single CABG procedures. [31] Furthermore, this study followed the full KDIGO criteria, which are objective and traceable measures, performed per usual care, and analyzed meticulously by observing hourly urine output and creatinine changes. When AKI was defined using only creatinine levels, the incidence in the control group dropped to 27 % in the control group, consistent with previously reported findings. [32] Although the AKI incidence decreased when defined solely by creatinine levels, the differences between the SGLT2 inhibition and control group were comparable.

A central hypothesized mechanism for the kidney protective effects of SGLT2 inhibitors is reducing metabolic workload and improving tissue oxygenation. To estimate the effect of empagliflozin on kidney hypoxia responses, we measured HIF-1α levels, an established biomarker for hypoxia and therapeutic target of SGLT2 inhibition. [33,34] Within this study, serum HIF-1α levels remained stable in the empagliflozin group, whereas HIF-1α increased in the control group after surgery. This aligns with earlier studies reporting that SGLT2 inhibition suppressed the HIF-1α pathway. [33,35] We think that the stable levels of HIF-1α after SGLT2 inhibition reflect the reduced hypoxic state in the treated kidneys and therefore lesser need of HIF-1α upregulation, which is positive given that prolonged HIF-1α activation is found to be detrimental, promoting inflammation and fibrosis. [35,36] Additionally, it has been shown (in mice) that SGLT2 inhibition after ischemia/reperfusion injury decreased endothelial thinning, hypoxia, and the development of kidney fibrosis. [37]

We initiated SGLT2 inhibition three days preoperatively, which contrasts with the current recommendation to withhold SGLT2

Table 2
Intraoperative and postoperative glycemic control and insulin administrations.

	Empagliflozin (n = 25)	Control (n = 30)	Absolute difference	95 % CI	P value
Intraoperative					
Mean blood glucose, mean ± SD, mmol/l	6.6 ± 1.5	7.1 ± 1.9	0.5	−0.3–1.2	0.21
Hyperglycemia (>10 mmol/l), n (%)	3 (12)	6 (20)	8	−27.2–11.2	0.487
Hypoglycemia (<4 mmol/l), n (%)	0 (0)	1 (3)	3	−9.8–3.1	1.000
Insulin administration, n (%)	1 (4)	2 (7)	3	−14.4–9.1	1.000
Postoperative[a]					
Peak blood glucose, mean ± SD, mmol/l	9.3 ± 1.5	10.9 ± 2.5	1.6	0.5–2.8	0.007
Hyperglycemia (>10 mmol/l), n (%)	7 (28)	18 (60)	32	−56.8 to −7.2	0.029
Hypoglycemia (<4 mmol/l), n (%)	0 (0)	0 (0)	NA	NA	NA
Insulin administration, n (%)	2 (8)	11 (37)	29	−48.9 to −8.4	0.023

Abbreviations: CI = confidence interval; NA = not applicable; SD = standard deviation.
[a] The first 48 h postoperative.

inhibitors three days before surgery to attenuate the associated risk of ketoacidosis. [38] In this study, ketone concentrations increased during cardiac surgery, as we reported previously. [19] However, we found no additional effect of SGLT2 inhibition and recorded no cases of ketoacidosis. Although the absence of ketoacidosis in this small pilot study does not prove safety, given its rare incidence and the low likelihood of detecting a case, this is reassuring for further research. Of note, no patients with insulin-dependent T2D were included in this study by chance, although the per-protocol administration of glucose/insulin infusion would have mitigated their risk of ketoacidosis. Furthermore, we found no between-group differences in the incidence of hypoglycemia or other adverse events. As expected from a glucose-lowering drug, the incidence of hyperglycemia was reduced in the SGLT2 inhibitor group. [39]

By design, this pilot study was limited by a small sample size and a (standard care) open-label control group. Therefore, chance may have affected the between-group differences in outcomes, and results should only be interpreted as hypothesis-generating. Our goals for this pilot study were achieved as the intervention was feasible and safe. We believed that the association between SGLT2 inhibition and a reduction in AKI incidence needed proper evaluation in a larger, placebo-controlled trial, which motivated the early termination of this pilot trial. Since then, our group has set up the multi-center MERCURI-2 trial. This study will test the hypothesis that SGLT2 inhibition reduces CSA-AKI in 784 patients undergoing cardiac surgery, with enrolment expected to be completed by the first quarter of 2025 (clinicaltrials.gov: NCT05590143).

5. Conclusion

In this pilot study, SGLT2 inhibition was not associated with a significant between-group difference in NGAL levels. We observed that it was feasible to start SGLT2 inhibition before cardiac surgery and found no effect of the intervention on intraoperative ketogenesis in this small study population. The observed reduced incidence of AKI according to KDIGO criteria informed the design and hypothesis of a large, placebo-controlled trial that has now been initiated.

CRediT authorship contribution statement

Lars I.P. Snel: Writing – original draft, Validation, Investigation, Formal analysis. **Maartina J.P. Oosterom-Eijmael:** Writing – original draft, Validation, Investigation, Formal analysis. **Elena Rampanelli:** Validation, Resources, Formal analysis. **Yugeesh R. Lankadeva:** Writing – review & editing, Supervision. **Mark P. Plummer:** Writing – review & editing. **Benedikt Preckel:** Writing – review & editing, Supervision, Resources. **Jeroen Hermanides:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Daniel H. van Raalte:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Abraham H. Hulst:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

DHR has served as a consultant and received honoraria from Boehringer Ingelheim and Lilly, Merck, Novo Nordisk, Sanofi, and AstraZeneca. He has also received research operating funds from Boehringer Ingelheim and Lilly Diabetes Alliance, AstraZeneca, and Novo Nordisk; all honoraria are paid to his employer (Amsterdam University Medical Center, Amsterdam). All other authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclinane.2025.111811.

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